Mavera click audit

Phase 1 of the onboarding and click-reduction plan. Measured 2026-10-22 with `tools/visual/click_audit.py`.

Baseline measured	git commit `9a087dc` (HEAD), archived with `git archive` and served on :5175 so the numbers stay reproducible
Live working tree	http://localhost:5173 at the same time, 114 files modified by concurrent work (command palette actions, provider account stores, assistant widget); where a live number differs it is shown as `baseline / now live`
Browser	Chromium headless 1440 x 900 (390 x 844 for the responsive sweep), signed in through the Demo mode card (button Owner)
Harness	`python tools/visual/click_audit.py` (see section 9), one fresh browser context per task, every task starts on the Dashboard

What is counted. An in-page listener counts only trusted events, so scripted `fill()` costs nothing and a real user cannot be under-counted:

- **Clicks**: mouse presses. A click that only focuses a text field is the first half of typing into it, so it is not added a second time.
- **Fields**: text fields typed into (one entry = focus + typing, counted once). **Keys**: non-printable key presses (Tab, Enter, Escape, arrows, chords such as Ctrl+K).
- **Interactions** = clicks (excluding field-focus clicks) + fields + keys. This is the number the checklist compares with the 3-interaction target. Typed characters are listed separately because they cannot be removed by layout alone.
- An opened `Select` costs 2 interactions (open + option). File pickers cost the click that opens them; picking the file in the OS dialog is not counted.

1. Connect flows (the owner's priority)

Each row starts on the Dashboard and ends with the account saved, using format-valid sample values (Azure UUIDs and a secret, `AKIA` + 16 characters and a 40-character secret, a service-account JSON, BTP URLs). Success is asserted in the page (confirmation text or the new row).

Flow	Interactions	Breakdown	Typed chars	Clicks that are pure overhead	Success feedback	Defects found while measuring
Connect Azure	**13**	8 clicks + 5 fields (menu 2, Manage Accounts, Connect Account, Next, Validate details, Save account, Done)	131	4 (Manage Accounts, Validate details screen, Done, and the Next/Save split)	wizard step 'Account connected'	At baseline the new account cannot be opened (measured: the cost page only knows fixture accounts, 'Account not found'); the live tree opens it. The overview never lists it and the account is lost when the manager unmounts (code)
Connect AWS	**11**	8 clicks + 3 fields (same four-step wizard, 3 fields)	69	4	wizard step 'Account connected'	Region is hard-coded to us-east-1 (not asked); overview row shows a dash for spend
Connect GCP	**7**	4 clicks + 3 fields (menu 2, Manage, 3 fields, Connect Project; inline form, no wizard)	89	1 (Manage)	inline notice for 4 seconds (not a toast)	Different UX from Azure/AWS; the pasted JSON is validated then discarded; 'Project ID' retyped although the JSON already contains it

Flow	Interactions	Breakdown	Typed chars	Clicks that are pure overhead	Success feedback	Defects found while measuring
Connect BTP	**9**	4 clicks + 5 fields (menu 2, Add Account, 5 fields, Connect Account; inline form)	113	1	none (form clears, row appears)	Region saved as `reporting` (first host label of the URL) instead of `eu10`; 5 required fields, none with an example for Client ID/Secret
Add SAP destination (menu path)	**7**	4 clicks + 3 fields (menu 2, Add Destination, 3 fields, submit)	39	0	toast 'added, ready to connect'	No test-connection step: 'Ready To Connect' is a label, the password is collected and thrown away, the destination switcher never lists the new system
Add SAP system from Dashboard Quick Access	**8 / now 5**	the Quick Access 'Add SAP System' link opens SAP Overview, which has no add control, so the user must start again from the menu	39	2 (wrong landing + re-navigation)	same toast	At baseline the link points at `/sap/systems` instead of `/sap/configuration`; the live tree sends it to System Configuration
Cost Overview 'Connect cloud account'	**11 (did not complete) / now 12**	menu 2, Connect cloud account, 5 Azure fields, Next, Validate, Save	132	3	fails in demo mode	This page mounts the wizard without the demo `onSave`, so Save posts to a real API and errors; no provider picker (Azure only)

What a 'new organisation' user faces today: five different connect UIs (wizard, two inline forms, destination dialog, Cost Overview wizard), none reachable from the Dashboard, none with a 'use sample values' path, and the two wizard providers spend 4 clicks on screens that carry no input.

2. Headlines

- Every menu page is 2 clicks from the Dashboard (measured for 31 destinations); account detail pages are 3, a tab is 3, a non-default view is 4.
- Three signed-in pages are not in the menu for any demo account: `/invoices` (4 clicks via Cost Overview > Invoices summary > link), `/alerts` (1 click from a Dashboard link, never from the menus) and `/ai-assistant` (Ctrl+K, type, arrow, Enter = 4; the widget is the intended entry).
- The expensive tasks are the connect flows (7 to 13 interactions), add user with a role (8), editing permissions of an existing custom role (7) and security-notes upload plus status check (6).
- Several pages are dead ends for the primary job: SAP Overview (no drill-in from a critical system), BTP Security (empty tables, dead Refresh, no data path), `/alerts` (static three-row table with no actions), Reserved Instances (read-only).
- Every page puts 9 header controls (logo, Dashboard, 3 menus, search, bell, help, profile) before the content. A 'Skip to content' link exists and works, so the keyboard cost is 2 keys, but nothing tells a sighted keyboard user.
- The command palette (built in the working tree, not at the baseline commit) shortens the connect launch from 4 clicks to 3 interactions and removes the hidden-page detour for invoice upload (7 to 5), but does not touch the form overhead inside the Azure and AWS wizards (section 6).
- No page keeps its state in the URL except hash tabs: spend period, Executive/Operations/Technical, alert filters and connect progress reset on every visit.

3. Measured tasks

Interactions from the Dashboard. 'Breakdown' is clicks + fields + keys. Where the working tree already differs from the baseline, the live number follows `/ now`.

Task	Group	Interactions	Breakdown	Typed chars	Status	Note
Open an Azure account detail (/azure/:id)	account-detail	**3**	3 clicks	0	ok	
Open an AWS account detail (/aws/account/:id)	account-detail	**3**	3 clicks	0	ok	
Open a GCP project detail (/gcp/account/:id)	account-detail	**3**	3 clicks	0	ok	
Open a BTP account detail (/btp/account/:id)	account-detail	**3**	3 clicks	0	ok	
AZURE account: switch to the Operations view	account-view	**4**	4 clicks	0	ok	
AZURE account: switch to the Technical view	account-view	**4**	4 clicks	0	ok	
AWS account: switch to the Operations view	account-view	**4**	4 clicks	0	ok	
AWS account: switch to the Technical view	account-view	**4**	4 clicks	0	ok	
GCP account: switch to the Operations view	account-view	**4**	4 clicks	0	ok	
GCP account: switch to the Technical view	account-view	**4**	4 clicks	0	ok	
BTP account: switch to the Operations view	account-view	**4**	4 clicks	0	ok	
BTP account: switch to the Technical view	account-view	**4**	4 clicks	0	ok	
BTP account: switch to the Hierarchy view	account-view	**4**	4 clicks	0	ok	
Cost Overview: change the spend period (6M to 3M)	cost	**3**	3 clicks	0	ok	
Cost Analytics: export a report (CSV)	cost	**3**	3 clicks	0	ok	
Cost Analytics: save the current view	cost	**3**	3 clicks	0	ok	
Cost Overview: 'Connect cloud account' wizard (Azure sample values)	connect	**11 (did not complete) / now 12**	6 clicks + 5 fields	132	failed	This page's wizard has no provider picker and posts to the real API (no demo onSave); expect failure at Save in demo mode. / TimeoutError: Locator.wait_for: Timeout 9000m

Task	Group	Interactions	Breakdown	Typed chars	Status	Note
Savings & Optimization: open a recommendation and start review	cost	**5**	5 clicks	0	ok	
Create a budget (name, amount, defaults for the rest)	governance	**5**	3 clicks + 2 fields	12	ok	
Add a notification channel (email)	governance	**6**	5 clicks + 1 field	15	ok	
Smart Alerts: resolve a cloud alert from the list	alerts	**3**	3 clicks	0	ok	
Smart Alerts: ignore an alert	alerts	**3**	3 clicks	0	ok	
Smart Alerts: open an alert's detail	alerts	**3**	3 clicks	0	ok	
Dashboard 'Needs attention': get to the alert that needs action and resolve it	alerts	**2**	2 clicks	0	ok	
Bell: mark all notifications read	alerts	**2**	2 clicks	0	ok	
SAP Alerts: resolve an alert from the list	alerts	**3**	3 clicks	0	ok	
Smart Alerts: add an alert recipient	alerts	**5**	4 clicks + 1 field	25	ok	
Upload an invoice (small dummy PDF) and see the analysis	invoices	**3**	3 clicks	0	ok	Starts at /invoices (reached via Cost Overview; 3 extra clicks counted separately in reach.invoices-via-overview).
Delete an invoice (select, trash, confirm)	invoices	**3**	3 clicks	0	ok	
Billing: pay the newest Sent invoice	invoices	**4**	4 clicks	0	ok	
Security Notes: upload note numbers by typing (3 notes)	sap	**5**	4 clicks + 1 field	25	ok	
Security Notes: upload 3 numbers AND resolve their status (Upload + Check Status)	sap	**6**	5 clicks + 1 field	25	ok	
Security Notes: run Check Status on the existing notes	sap	**3**	3 clicks	0	ok	
Security Notes: delete a note (trash + confirm)	sap	**4**	4 clicks	0	ok	
SAP: add a destination / system via System Configuration (name, SID, URL)	connect	**7**	4 clicks + 3 fields	39	ok	
SAP: add a system starting from Dashboard Quick Access 'Add SAP System'	connect	**8 / now 5**	5 clicks + 3 fields	39	ok	The link lands on /sap/systems, which has no add flow; the user must then go to System Configuration through the menu.

Task	Group	Interactions	Breakdown	Typed chars	Status	Note
Jobs & Dumps: filter by status (Running) and search	sap	**4**	3 clicks + 1 field	3	ok	
HANA: open backup details (Backups tab, select a marker)	sap	**4**	4 clicks	0	ok	
Transport: import all released requests	sap	**4**	4 clicks	0	ok	
SAP Overview: open the critical system (HANA-PRD) for detail	sap	**3 (did not complete)**	3 clicks	0	failed	No drill-in exists / TimeoutError: Locator.wait_for: Timeout 6000ms exceeded.
Add a user with a role (Analyst) and no extra access	users	**8**	6 clicks + 2 fields	36	ok	
Add a user keeping the default role (Viewer): name + email + Enter	users	**7**	3 clicks + 2 fields + 2 keys	36	ok	
Change a member's role (row Select)	users	**4**	4 clicks	0	ok	
Add a custom role (name only)	users	**6**	4 clicks + 1 field + 1 key	10	ok	
Duplicate a role (row action)	users	**4**	4 clicks	0	ok	
Edit a custom role's permissions: grant 'All' on one module and save	users	**8**	6 clicks + 1 field + 1 key	10	ok	Setup (creating the custom role) is not counted; path starts at the dashboard and ends with Save changes.
Edit permissions of an EXISTING custom role (tab, role Select, area/module, All, Save)	users	**7**	7 clicks	0	ok	
Audit Trail: open an event	users	**3**	3 clicks	0	ok	
Sync Logs: filter by status (Failed)	users	**3**	3 clicks	0	ok	
Subscription: request a plan	users	**5**	5 clicks	0	ok	
Account: switch to dark theme	users	**2**	2 clicks	0	ok	
Log out (profile menu, Log out, confirm)	users	**3**	3 clicks	0	ok	
Connect Azure (wizard): menu, Manage Accounts, Connect Account, 5 fields, Next, Validate, Save, Done	connect	**13**	8 clicks + 5 fields	131	ok	
Connect Azure then OPEN the new account	connect	**14 (did not complete) / now 14**	9 clicks + 5 fields	131	failed	Known defect: the new Azure account cannot be opened (cost page only knows the fixtures). / TimeoutError: Locator.wait_for: Timeout 3000ms exceeded.

Task	Group	Interactions	Breakdown	Typed chars	Status	Note
Connect AWS (wizard): menu, Manage Accounts, Connect Account, 3 fields, Next, Validate, Save, Done	connect	**11**	8 clicks + 3 fields	69	ok	
Connect GCP (inline form): menu, Manage, 3 fields, Connect Project	connect	**7**	4 clicks + 3 fields	89	ok	
Connect BTP (inline form): menu, Add Account, 5 fields, Connect Account	connect	**9**	4 clicks + 5 fields	113	ok	
Ask the assistant a question via a suggestion chip	assistant	**2**	2 clicks	0	ok	
Ask the assistant a typed question	assistant	**3**	1 click + 1 field + 1 key	35	ok	
Start a guided tour of the page (first-time user: chip in the launcher panel)	assistant	**2**	2 clicks	0	ok	
Start a guided tour when the 'New here?' prompt is gone (launcher, Tour, pick)	assistant	**3**	3 clicks	0	ok	
Ctrl+K to reach a page (type 'budg', Enter)	assistant	**3**	0 clicks + 1 field + 2 keys	4	ok	
Ctrl+K: run an ACTION ('connect azure') rather than open a page	assistant	**2 (did not complete) / now 3**	0 clicks + 1 field + 1 key	13	failed	TimeoutError: Locator.wait_for: Timeout 6000ms exceeded.
Admin: reach Organizations	super-admin	**1**	1 click	0	ok	
Admin: reach Billing	super-admin	**1**	1 click	0	ok	
Admin: open the new-organization form	super-admin	**2**	2 clicks	0	ok	
Admin: create an organization (name, 1 module, 1 provider)	super-admin	**6**	5 clicks + 1 field	16	ok	
Admin: open an organization's settings	super-admin	**2**	2 clicks	0	ok	
Admin: open the invoice generator from the dashboard	super-admin	**1**	1 click	0	ok	

Task	Group	Interactions	Breakdown	Typed chars	Status	Note
Admin: create and send an invoice (1 line)	super-admin	**7**	5 clicks + 2 fields	25	ok	
Admin: reach Customers (Billing, Manage Customers)	super-admin	**2**	2 clicks	0	ok	
Resource Inventory: open a resource's detail	cost	**3**	3 clicks	0	ok	
Resource Inventory: show only untagged resources (Tags filter)	cost	**4**	4 clicks	0	ok	
Reserved Instances: pick a reservation to chart	cost	**3**	3 clicks	0	ok	
Palette: Connect Azure (Ctrl+K, type, Enter, then the wizard)	palette	**not available / now 12**	0 clicks + 1 field + 1 key	13	failed	AssertionError: Ctrl+K has no action 'Connect Azure' (page search only)
Palette: Connect AWS (Ctrl+K, type, Enter, then the wizard)	palette	**not available / now 10**	0 clicks + 1 field + 1 key	11	failed	AssertionError: Ctrl+K has no action 'Connect AWS' (page search only)
Palette: Connect Google Cloud project (inline form, first field focused)	palette	**not available / now 7**	0 clicks + 1 field + 1 key	14	failed	AssertionError: Ctrl+K has no action 'Connect Google Cloud' (page search only)
Palette: Connect SAP BTP account (inline form, first field focused)	palette	**not available / now 9**	0 clicks + 1 field + 1 key	15	failed	AssertionError: Ctrl+K has no action 'Connect SAP BTP' (page search only)
Palette: Connect SAP system (destination dialog opens, Environment Name focused)	palette	**not available / now 7**	0 clicks + 1 field + 1 key	18	failed	AssertionError: Ctrl+K has no action 'Connect SAP system' (page search only)
Palette: Add budget (opens Budgets with Budget Name focused)	palette	**not available / now 6**	0 clicks + 1 field + 1 key	10	failed	AssertionError: Ctrl+K has no action 'Add budget' (page search only)
Palette: Upload invoice (opens the upload dialog on /invoices)	palette	**not available / now 5**	0 clicks + 1 field + 1 key	14	failed	AssertionError: Ctrl+K has no action 'Upload invoice' (page search only)
Palette: Add user (opens the dialog, Full name focused; default role)	palette	**not available / now 7**	0 clicks + 1 field + 1 key	8	failed	AssertionError: Ctrl+K has no action 'Add user' (page search only)
Palette: Upload security notes (dialog opens with the textarea focused)	palette	**not available / now 5**	0 clicks + 1 field + 1 key	21	failed	AssertionError: Ctrl+K has no action 'Upload security notes' (page search only)

4. Reaching every page

Destination	Path	Interactions
Dashboard (/)	click header link 'Dashboard'	1
SAP Overview (/sap/systems)	click header menu 'SAP Monitoring'; click menu item 'SAP Overview'	2
HANA/DB Monitoring (/sap/databases)	click header menu 'SAP Monitoring'; click menu item 'HANA/DB Monitoring'	2
Performance (/sap/performance)	click header menu 'SAP Monitoring'; click menu item 'Performance'	2
Jobs & Dumps (/sap/jobs)	click header menu 'SAP Monitoring'; click menu item 'Jobs & Dumps'	2
Alerts & Escalations (/sap/alerts)	click header menu 'SAP Monitoring'; click menu item 'Alerts & Escalations'	2
Transport Management (/sap/transport-management)	click header menu 'SAP Monitoring'; click menu item 'Transport Management'	2
Security Notes (/sap/security-notes)	click header menu 'SAP Monitoring'; click menu item 'Security Notes'	2
BTP Security Assessment (/sap/btp-security)	click header menu 'SAP Monitoring'; click menu item 'BTP Security Assessment'	2
System Configuration (/sap/configuration)	click header menu 'SAP Monitoring'; click menu item 'System Configuration'	2
Cost Overview (/cloud/overview)	click header menu 'Cloud Cost'; click menu item 'Cost Overview'	2
Cost Analytics (/cloud/analytics)	click header menu 'Cloud Cost'; click menu item 'Cost Analytics'	2
Azure Subscriptions (/azure)	click header menu 'Cloud Cost'; click menu item 'Azure Subscriptions'	2
AWS (/aws)	click header menu 'Cloud Cost'; click menu item 'AWS'	2
SAP BTP (/btp)	click header menu 'Cloud Cost'; click menu item 'SAP BTP'	2
Google Cloud (/gcp)	click header menu 'Cloud Cost'; click menu item 'Google Cloud'	2
Savings & Optimization (/cloud/optimization)	click header menu 'Cloud Cost'; click menu item 'Savings & Optimization'	2
Reserved Instances (/cloud/reservations)	click header menu 'Cloud Cost'; click menu item 'Reserved Instances'	2
Resource Inventory (/cloud/resources)	click header menu 'Cloud Cost'; click menu item 'Resource Inventory'	2
Budgets & Alerts (/budgets)	click header menu 'Cloud Cost'; click menu item 'Budgets & Alerts'	2
Smart Alerts (/smart-alerts)	click header menu 'Cloud Cost'; click menu item 'Smart Alerts'	2
Billing (/billing)	click header menu 'Cloud Cost'; click menu item 'Billing'	2
User Management (/users)	click header menu 'Administration'; click menu item 'User Management'	2
Audit Trail (/audit-trail)	click header menu 'Administration'; click menu item 'Audit Trail'	2
Sync Logs (/sync-logs)	click header menu 'Administration'; click menu item 'Sync Logs'	2

Destination	Path	Interactions
/invoices via the Cloud Cost menu (menu entry exists for invoice-only orgs only)	no menu entry for any demo account (needs an invoice-only or hybrid organisation)	not reachable by menu; see the Cost Overview detour below
/alerts (not in the menu) from the dashboard	click dashboard link to /alerts	1
/ai-assistant (not in the menus or the dashboard)	press Ctrl+K; type 'assistant' (9 chars, field auto-focused); press ArrowDown to the 'AI Assistant' page result; press Enter	4
/account via the profile menu	click Profile menu; click Account	2
/subscription via the profile menu	click Profile menu; click Subscription	2
/invoices when it is not in the menu (Cost Overview > Invoices summary > Upload more invoices)	click header menu 'Cloud Cost'; click menu item 'Cost Overview'; click 'Invoices summary' toggle; click link to Invoices	4
Open the floating assistant panel from any page	click assistant launcher	1

Tabs and views reachable by hash or switcher:

Tab or view	Interactions	Hash
Reach tab 'Services and disk' on HANA/DB Monitoring (#storage)	3	
Reach tab 'Backups' on HANA/DB Monitoring (#backups)	3	
Reach tab 'Jobs and processes' on Performance (#jobs)	3	
Reach tab 'Analytics' on Performance (#analytics)	3	
Reach tab 'Dumps' on Jobs & Dumps (#dumps)	3	
Reach tab 'Settings' on Alerts & Escalations (#settings)	3	
Reach tab 'Recipients' on Alerts & Escalations (#recipients)	3	
Reach tab 'Alert integrations' on Budgets & Alerts (#integrations)	3	
Reach tab 'Recipients' on Smart Alerts (#recipients)	3	
Reach tab 'Settings' on Smart Alerts (#settings)	3	
Reach tab 'Opportunities' on Savings & Optimization (#opportunities)	3	
Reach tab 'Roles' on User Management (#roles)	3	
Reach tab 'Permissions' on User Management (#permissions)	3	
Reach tab 'Available plans' on /subscription (#plans)	3	
Reach tab 'Security' on /account (#security)	3	
Reach tab 'Appearance' on /account (#appearance)	3	
Reach tab 'Help' on /account (#help)	3	

5. Per-route checklist

Each cell is `pass` or `fail` followed by the measured number or the code evidence. **Reach** counts interactions from the Dashboard (a menu page is 2). **Primary task** counts interactions on the page itself: the measured total from the Dashboard minus the interactions spent reaching it (2 for menu pages, 3 for account detail pages), against a target of 3 or fewer. **Empty state**, **keyboard** and **shared components** come from reading the page and its section components. **Deep links** were exercised in the browser (a hash or path loaded straight into the right tab or view) and the rest is from code. **Responsive** was measured at 1440 and 390 px: a page that scrolls sideways or has content wider than the viewport fails; a table that scrolls inside its own wrapper is noted (the owner rule is no inner scrollbars, listed as a cross-cutting finding) but does not fail this column. 'n/a' means the page has no empty state to speak of or no demo data path exists.

Keyboard column quotes the measured Tab presses: every signed-in page has the same header, so the first Tab lands on the 'Skip to content' link (which works), then 9 header controls, then the page.

5.1 Dashboard and cloud cost

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/` Dashboard	pass: **1** (Dashboard link from any page)	pass: act on a cost alert from 'Needs attention' (row, Resolve): **2**	fail: 'Nothing needs attention right now.' is plain text; with zero accounts the KPIs show a dash and nothing says how to connect	pass: no page state to link (Time range, Environment, System are remembered per session but filter nothing)	pass: Needs-attention rows are real links; KPI cards are not focusable [11 Tab presses to the first control in the page]	pass	pass: SearchField, ModuleCard, SapSystemsPanel reused. Defect: the search box and 3 Selects are decorative (their state is never read)
`/cloud/overview` Cost Overview	pass: **2**	pass: change the spend period: **1**	fail: no cloud access shows 'Connect your first cloud account' with no button; other cards give text only	fail: period (default 6M) and the Cloud/Invoices toggle are plain state: not in the URL, not remembered	pass: real buttons and links; but 6 plain `<a href>` links reload the whole app and wipe in-memory demo state [11 Tab presses to the first control in the page]	pass	fail: 6 hard `<a href>` instead of `Link`; its Connect wizard is mounted without the demo onSave; hand-written KPI shells
`/cloud/analytics` Cost Analytics	pass: **2**	pass: export a report: **1**	fail: 'No service data for this scope.' is text only	fail: scope, range, group live in sessionStorage; saved views in localStorage (max 6, no delete); none in the URL	fail: the line chart has no keyboard or data-table alternative; breakdown rows open only through the name button [11 Tab presses to the first control in the page]	pass: inner scroller(s) 1440: y:mv-table-wrap	fail: own `div.mv-filter-bar` (not FilterBar), saved-view chips are bare `button.mv-chip` (not FilterChips)

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/cloud/optimization` Savings & Optimization	pass: **2** (#opportunities tab: 3)	pass: open a recommendation and start review: **3**	fail: 'No opportunities match these filters.' has no Clear filters button	pass: #overview and #opportunities load the right tab (measured); filters only in sessionStorage	pass: TabBar arrows, drawer Escape and focus return. Defect: the floating assistant launcher covers the drawer's 'Start review' button, which had to be clicked at its left edge [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: own filter bar div, no FilterChips; assigning an owner needs a drawer per row (no bulk)
`/cloud/reservations` Reserved Instances	pass: **2**	pass: pick a reservation to chart: **1**	fail: 'No reservations or savings plans for this provider.' text only	fail: provider and selected reservation are sessionStorage only	fail: the utilisation line chart has no data-table alternative [11 Tab presses to the first control in the page]	pass	fail: own table, no ClickableRow; read-only page (no export or renew action)
`/cloud/resources` Resource Inventory	pass: **2**	pass: open a resource: **1**	pass: 'No resources match these filters.' with a Clear filters button	fail: all six filters in sessionStorage; drawer not in the URL	pass: name button opens the drawer; focus returns on close [11 Tab presses to the first control in the page]	pass: inner scroller(s) 1440: y:mv-table-wrap	pass: FilterBar, SearchField, DetailDrawer, CopyButton (renders every row, no Pagination, vertical inner scroller at 1440)
`/budgets` Budgets & Alerts	pass: **2** (#integrations tab: 3)	pass: create a budget (name, amount, defaults): **3**	pass: 'Add your first channel' button; the budget list says 'No Budgets Found' beside the create form	pass: #budgets and #integrations (measured); the create draft survives a tab switch	fail: delete budget and remove channel use ConfirmProvider (no Escape, no focus move, no trap); the Add channel dialog opens focused on 'Close' [11 Tab presses to the first control in the page]	pass	pass: TabBar, useHashTab, ConfirmProvider, useModalOverlay dialog
`/smart-alerts` Smart Alerts	pass: **2** (#recipients and #settings tabs: 3)	pass: resolve an alert from the list: **1**	pass: 'Show active alerts' button when filtered; 'All clear' otherwise	fail: tabs by hash (measured) but status/type filters and the open drawer are not in the URL or remembered	pass: every action is a real button with a row-specific aria-label [11 Tab presses to the first control in the page]	pass	fail: status and type are SegmentedToggle without counts (FilterChips fit), own `div.mv-cost-alert-filters`, no ClickableRow

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/alerts` Alerts (in no menu)	pass: **1** (only via a Dashboard link)	fail: no task is possible: static 3-row table, no resolve/ignore/open, no link to the owning module	fail: 'No active alerts.' text only	n/a: no state	fail: no focusable control inside the page [no focusable control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: hand-written table; its data is a fixture separate from Smart Alerts, SAP Alerts and the bell
`/invoices` Invoices (menu entry hidden for every demo account)	fail: **4** (Cost Overview, Invoices summary toggle, link). It is not in the Cloud Cost menu: that needs an invoice-only or hybrid organisation	pass: upload an invoice and see the analysis: **3**	pass: 'Upload Invoice' button (the wording 'No invoices yet' also shows for an empty provider filter)	fail: selected invoice and page are local state	fail: UploadModal has no Escape, no focus trap, no autofocus; a backdrop click closes it mid-parse [11 Tab presses to the first control in the page]	pass	fail: hand-written modal (not useModalOverlay), `div.mv-empty`, KPI articles, invoice cards
`/billing` Billing	pass: **2**	pass: pay the unpaid invoice (select it, Pay Now): **2**	fail: 'No invoices yet' with no action	fail: selected invoice is local state	pass: list items are buttons; Pay Now is one click with no confirmation (risk, not a keyboard gap) [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: hand-written invoice list and empty states; only PageHeader is shared
`/ai-assistant` AI Assistant (page)	fail: **4** (Ctrl+K, type, ArrowDown past the 'Open assistant' action, Enter); in no menu	pass: ask a question: 3 on the page (composer, type, Enter), 2 with a suggestion chip; measured in the floating widget: 3 typed, 2 with a chip	pass: intro line and suggestion chips	n/a: the conversation is not persisted	pass: form submit on Enter; the widget composer is not auto-focused [11 Tab presses to the first control in the page]	pass	fail: the page and the floating widget each implement a conversation with the same input label and separate state

5.2 Cloud providers

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/azure` Azure overview and account manager	pass: **2**	fail: connect an Azure account: **11**	fail: zero accounts: the manager shows only 'Connect Account' with no 'no accounts' message; the overview never has an empty state	fail: overview vs manager is a local switch; `/azure` is the only URL for both	fail: the wizard has no Escape, trap or autofocus and no `<form>` (Enter does nothing); header dropdowns have no Escape or arrows [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: no SearchField, FilterBar, TabBar, EmptyState or StatusTag; the account-manager markup is copied across Azure, AWS and BTP
`/azure/:accountId` Azure account (Executive, Operations, Technical)	pass: **3** (menu, list, row; fixed 3)	pass: switch to Operations or Technical: **1**	fail: 'Account not found' + Back; empty panels are text only ('No data available yet.')	fail: `/azure/az-acc-1` loads (measured) but the view and the month are not in the URL or remembered	fail: the view switcher is a `nav` of buttons with aria-current, no tab roles or arrows; the month dropdown has no Escape [11 Tab presses to the first control in the page]	pass	fail: own `mv-cost-tabs` (4 copies), own dropdown + useOutsideClose (4 copies), Sparkline re-implemented, formatCurrency declared in ~15 files
`/aws` AWS overview	pass: **2**	pass: open an account: **1**	fail: zero accounts renders an empty table and '0 AWS environments' with no action	pass: its own URL	pass: ClickableRow + RowOpenButton (whole row, Enter) [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: Sparkline re-implemented; static Tag Coverage and Status columns
`/aws/accounts` AWS account manager	fail: **3** (menu 2 + Manage Accounts)	fail: connect an AWS account: **9**	fail: zero accounts: only the notice and Connect Account	pass: its own URL	fail: same wizard gaps as Azure [11 Tab presses to the first control in the page]	pass	fail: duplicate of the Azure manager and wizard
`/aws/account/:id` AWS account	pass: **3** (fixed 3)	pass: switch view: **1**	fail: 'Account not found' + Back	fail: view and month not in the URL	fail: cost-view buttons without tab roles [11 Tab presses to the first control in the page]	pass	fail: copy of the Azure cost-page shell
`/gcp` GCP overview	pass: **2**	pass: open a project: **1**	pass: 'No GCP projects connected' + 'Connect GCP Project' (the only provider with a real zero state)	pass: its own URL	pass: project rows are real buttons [11 Tab presses to the first control in the page]	pass	fail: no ClickableRow or table; hand-written account rows
`/gcp/accounts` GCP account manager	fail: **3** (menu 2 + Manage)	fail: connect a GCP project: **5**	pass: the connect form is always on the page	pass: its own URL	pass: a real `<form>`: Enter submits; field errors are inline [11 Tab presses to the first control in the page]	pass	fail: a third connect UI (inline) unlike Azure/AWS; the assistant flow-guide targets (`Rename`, `Delete`, `Connect Account`) do not exist here

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/gcp/account/:id` GCP project	pass: **3** (fixed 3)	pass: switch view: **1**	fail: 'Project not found' + Back	fail: view not in the URL; no month picker	fail: cost-view buttons without tab roles [11 Tab presses to the first control in the page]	pass	fail: own 5-KPI row and tables instead of CloudSpendSummary, ServiceBreakdownList, DailyCostHistory, CostByRegion
`/btp` SAP BTP overview	pass: **2**	pass: open an account: **1**	pass: 'No BTP accounts added yet' + 'Add your first account' (the rest of the page still shows fixture data)	pass: its own URL	pass: account rows are buttons [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: 'Manage' and 'Add Account' go to the same page; hand-written rows
`/btp/accounts` BTP account manager	fail: **3** (menu 2 + Add Account)	fail: connect a BTP account: **7**	fail: zero accounts: the form only; at the 3-account limit the form is hidden (the demo seeds 2, so one add is possible)	pass: its own URL	pass: a real `<form>`; validation is a single role=alert [11 Tab presses to the first control in the page]	pass	fail: third connect UI; no success message
`/btp/account/:id` BTP account (Executive, Operations, Hierarchy, Technical)	pass: **3** (fixed 3)	pass: switch view: **1**	fail: 'Account not found'; the 'CIS_Central not configured' notice has no link to fix it	fail: view and hierarchy drill-down are local state (the browser Back button does not step out of the drill-down)	fail: cost-view buttons without tab roles [11 Tab presses to the first control in the page]	pass	fail: own tabs and dropdowns as the other cost pages
`/btp/account/:id/subaccount/:id/service-instances` BTP service instances	fail: 4 or more (account, Hierarchy tab, subaccount, link) or the URL	fail: no task: the Refresh button has no handler although the hint says 'click Refresh'	fail: 'Service Manager not configured' and 'No instances found' without an action	pass: full URL (a direct load leaves the Back button with no history)	fail: the Back button calls `navigate(-1)`	not swept (sub-account route not in the sweep)	fail: dead Refresh; hand-written table

5.3 SAP monitoring

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/sap/systems` SAP Overview	pass: **2**	fail: open the critical system for detail: **not possible** (measured: clicking the HANA-PRD row opens nothing; rows are not clickable and there is no drill-in)	fail: an empty system list renders an empty tbody, no state, no action	pass: its own URL (`/sap/health` redirects here)	fail: the system table and severity bars are not interactive, so a keyboard user can reach only 'View all' and 'Open Jobs & Dumps' [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: condition pills are hand-written (not FilterChips, not clickable); table rows are not ClickableRow
`/sap/databases` HANA/DB Monitoring	pass: **2** (#backups tab: 3)	pass: open backup details (Backups tab, View data): **2**	fail: panels render empty lanes or lists with no message	pass: #vitals, #storage, #backups (measured)	fail: backup markers are aria-hidden with tabindex -1 (hover-only); the keyboard path is the 'View data' table; Refresh gives no feedback [11 Tab presses to the first control in the page]	pass	fail: SAP-local SegmentedSwitch (a copy of SegmentedToggle) and three copies of the View data dialog
`/sap/performance` Performance	pass: **2**	pass: filter jobs by status from here: **1** (chip link to `/sap/jobs?status=...`, code count)	fail: bare text ('No locks held', 'No memory pools reported')	pass: #health, #jobs, #analytics (measured)	pass: TabBar arrows; chips are real links [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: SegmentedSwitch and ChartDataDialog duplicates
`/sap/jobs` Jobs & Dumps	pass: **2** (#dumps tab: 3)	pass: filter by status and search: **2**	pass: 'Clear filters' when nothing matches (the dumps tab EmptyState has no action)	fail: #jobs and #dumps work (measured) and `?status=` is read once, but chip and search are never written back to the URL	pass: search, chips (aria-pressed), job-name buttons; Escape closes the drawer [11 Tab presses to the first control in the page]	pass: inner scroller(s) 1440: y:mv-table-wrap; 390: mv-table-wrap	pass: SearchField, FilterBar, FilterChips, TabBar, DetailDrawer (a vertical inner scroller on the table at 1440)
`/sap/alerts` Alerts & Escalations	pass: **2** (#settings, #recipients tabs: 3)	pass: resolve an alert: **1**	pass: 'Show active alerts' button	fail: tabs by hash; source, status and type filters are local state	pass: row actions are labelled buttons [11 Tab presses to the first control in the page]	pass	fail: hand-written `mv-filter-bar` with two Selects (no FilterBar, FilterChips or SearchField); Reopen is reachable only after switching the Status filter
`/sap/transport-management` Transport Management	pass: **2**	pass: import all released requests (select all, Import Selected): **2**	fail: 'No transports found.' with no Clear filters	fail: search, status and selection are local	pass: native checkboxes; the status toggle needs a Tab per option (no arrows) [11 Tab presses to the first control in the page]	pass	fail: status filter is a SegmentedToggle where Jobs uses FilterChips; import has no confirmation

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/sap/security-notes` Security Notes	pass: **2**	fail: upload note numbers AND resolve their status (Upload + Check Status): **4**	pass: 'Upload notes' button when no notes are tracked	fail: priority/status/download filters remembered in sessionStorage (search is not); nothing in the URL	fail: the delete confirm (ConfirmProvider) has no Escape or focus handling; the upload dialog itself is fine (autofocus, Escape, trap) [11 Tab presses to the first control in the page]	pass	fail: status counts are KPI cards, not FilterChips; Upload and Check Status are two manual steps (upload alone: 3)
`/sap/btp-security` BTP Security Assessment	pass: **2**	fail: nothing works: the tables are empty by design, Refresh has no handler, search and the status filter filter nothing	fail: a 'No Users Found' row with no guidance (for example connect a BTP destination)	fail: the tab choice resets on reload	fail: hand-written tab buttons: no roles, no arrows, no hash [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: hand tabs (not TabBar), own filter bar, imports `src/demo` directly (architecture ratchet)
`/sap/configuration` System Configuration	pass: **2**	fail: add a destination (name, SID, URL): **5**	pass: 'Add Destination' button in the empty state	fail: dialog and search are local state	pass: dialog autofocus, Escape, Tab trap, Enter submits [11 Tab presses to the first control in the page]	pass	fail: empty block is `mv-empty-inline` (not EmptyState); no test-connection; DestinationSwitcher is hard-coded and ignores destinations

5.4 Administration and account

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/users` User Management	pass: **2** (#roles, #permissions tabs: 3)	fail: add a user (default role): **5**	pass: 'Add user' button in the empty member state	fail: tabs by hash (measured); the selected role in Permissions, the search and the unsaved permission draft are not (the draft is silently dropped on a tab switch)	pass: dialogs: autofocus, Escape, trap, Enter submits; the role Select is keyboard-operable [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: own switch Toggle and StatusBadge; rows are not ClickableRow and a member has no detail view
`/audit-trail` Audit Trail	pass: **2**	pass: open an event: **1**	pass: 'Clear filters' (the heading always names the actor filter even when another filter caused it)	fail: filters, page and the open event are local	fail: the actor `radiogroup` has no arrow-key handling; the row is not clickable, only the action text [11 Tab presses to the first control in the page]	pass	fail: hand-written actor radiogroup (not SegmentedToggle or FilterChips); no ClickableRow
`/sync-logs` Sync Logs	pass: **2**	pass: filter by status: **1**	pass: 'Clear filters'	fail: filters are local	fail: toolbar buttons expose no pressed state (no aria-pressed) [11 Tab presses to the first control in the page]	pass: inner scroller(s) 390: mv-table-wrap	fail: hand-written toolbar (no FilterBar, FilterChips or SearchField); imports `src/demo` directly

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/subscription` Subscription (profile menu)	pass: **2**	pass: request a plan: **3**	n/a: read-only page that always has data	pass: #plan and #plans (measured)	pass: dialog autofocus and Escape; Enter in the textarea adds a newline [11 Tab presses to the first control in the page]	pass	pass: TabBar, PageHeader; inline-styled rows
`/account` Account (profile menu)	pass: **2** (#security, #appearance, #help tabs: 3)	pass: switch theme: 2 from anywhere through the profile menu (measured: 2); 2 on the page (Appearance tab, option)	n/a	pass: #profile, #security, #appearance, #help (measured for #security)	fail: Theme and Density are hand-written radiogroups with no arrow keys [11 Tab presses to the first control in the page]	pass	fail: hand radiogroups; the Help tab holds one link; Density duplicates the Pagination footer toggle

5.5 Super admin portal (signed in as Super Admin)

Reach counts are from `/admin/dashboard`, the landing page.

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/admin/dashboard` Admin dashboard	pass: 0 (landing page)	pass: start a new invoice: **1**	fail: 'No invoices yet.' text only	pass: its own URL	pass: skip link and NavLinks; the menu drawer below 992 px has no focus trap	pass: inner scroller(s) 390: mv-table-wrap	fail: hand tables; 'daysLeft' hard-codes today as 2026-09-27
`/admin/organizations` Organizations	pass: **1**	pass: open an organization (Manage): **1**	fail: an empty list renders nothing	pass: its own URL	pass: Manage is a link per row; the rows themselves are not clickable	pass	fail: no SearchField, FilterBar, ClickableRow or Pagination
`/admin/organizations/new` New organization	pass: **2**	fail: create an organization (name, module, provider, Create): **4**	n/a: a form	pass: its own URL	fail: not a `<form>`: Enter does not save	pass	fail: hand-written sections; no duplicate-name check
`/admin/organizations/:orgId` Organization settings	pass: **2**	pass: save changes: 1 click (code count; the dirty guard is on Cancel only)	n/a: 'Organization not found' + link back	pass: direct URL	fail: no Enter-to-save; no guard on browser Back when dirty	not swept (needs an organization id)	fail: own Section panels
`/admin/billing` Billing (Invoices, Payment Terms)	pass: **1**	pass: new invoice: **1** (header button, code count)	fail: 'No invoices found' without a clear-filters action	fail: the Invoices/Payment Terms switch is local state	fail: invoice rows are `<tr onClick>` only (mouse-only); the tab pair has no ARIA	pass: inner scroller(s) 390: mv-table-wrap	fail: own search input (not SearchField), own segmented pair (not TabBar)

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/admin/billing/generate` Invoice generator	pass: 1 from the dashboard button, 2 from Billing	fail: create and send an invoice (1 line): **6**	n/a	pass: its own URL	fail: two confirmation modals without Escape or focus trap; `window.prompt` for a column label	pass: inner scroller(s) 390: mv-table-wrap	fail: own modals and `window.prompt`
`/admin/billing/customers` Customers	pass: **2**	pass: add a customer: 3 interactions (name, email, Add; code count)	fail: no empty state	pass: its own URL	pass: a real form	pass	fail: the list is page-local and does not feed the invoice generator; delete has no confirm

5.6 Public pages

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/login` Sign in (out of scope; demo card measured)	pass: 0 (entry)	pass: demo sign-in: **1** click on a Demo mode card button (it starts every task here); password sign-in untouched	n/a	pass: `?error=` banners	pass: autofocused email, Enter continues	pass	pass: PasswordField; Demo mode card is its own markup
`/register` Register (invitation)	fail: no inbound link from /login or /company-signup (measured: none found); URL only	fail: 5 steps; code count: email, code, 4 detail fields, 4 access boxes and 5 buttons, none is a `<form>` so Enter never submits	n/a	pass: its own URL	fail: not forms; no autofocus; errors in one alert with no focus move	pass	fail: hand-rolled strength text instead of PasswordStrengthMeter; overlaps /company-signup
`/company-signup` Company sign-up	pass: **1**	pass: first two steps (Use sample token, Validate license): **2**; the full flow is about 7 clicks + 4 entries (code count)	n/a	pass: its own URL (step state is not linkable)	pass: real forms; focus moves to the step heading; the first error is focused	pass	pass: PasswordStrengthMeter, Select, stepper
`/forgot-password` Forgot password	fail: **3** (needs an e-mail typed on /login first)	pass: email + submit: 2 (code count)	n/a	pass: its own URL	pass: a real form; no autofocus; the e-mail is not carried over from /login	pass	pass
`/reset-password` Reset password	n/a: only from the e-mailed link	pass: 2 password fields + submit: 3 (code count)	pass: a missing token shows 'Request a new link'	pass: `?token=`	pass: a real form; no strength meter	pass	pass

Route	Reach in <=2 clicks?	Primary task in <=3 interactions?	Empty state with a next action?	Deep-linkable?	Keyboard path?	Responsive (no overflow 390/1440)?	Shared components (no duplicates)?
`/subscription-expired` Subscription expired	n/a: forced redirect	pass: 1 click 'Sign out and use another account'; no in-app plan request here	n/a	pass: its own URL	pass: buttons and links	pass	fail: its own hard-coded PLANS list differs from the Subscription page's
`/invoice-demo` Public invoice parser	pass: **1**	pass: parse an invoice: **3** from /login (link, file, Parse invoice) = 2 on the page	n/a	pass: its own URL	pass: fieldset radios, focus management on the result	pass	fail: bespoke `mv-idemo-*` markup and a drop zone that /invoices (the signed-in upload) lacks
`/invoice/:token` Public invoice (e-mail link)	n/a: only from the invoice e-mail	pass: pay: 1 click, no confirmation	fail: 'Invoice not found' without an action	pass: token URL	pass: buttons and links	pass	fail: hard-coded light colours
`*` 404 page	n/a	pass: Back to dashboard: 1	pass: 'Back to dashboard'	n/a	pass: a link	pass	fail: hand-written empty markup, not EmptyState

Redirects: `/admin/login` to `/login`; `/invoice-builder` to `/admin/billing/generate`; `/recommendations` to `/cloud/optimization` (a hash is lost); `/reports` to `/cloud/analytics`; `/sap/health` to `/sap/systems`; `/sap/dumps` to `/sap/jobs#dumps`.

6. Command palette (already built in the working tree)

Ctrl+K now runs actions as well as opening pages. The baseline commit only searches pages (the harness confirms it: `Ctrl+K has no action 'Connect Azure'`). Measured on the live tree, each palette flow starts on the Dashboard and ends with the thing saved. Typing the query is one field entry, so Ctrl+K + query + Enter = 3 interactions (and 1 chord + N characters + 1 key on the keyboard), against 4 menu clicks to reach the Azure connect dialog.

Flow	Baseline (menu path) interactions	Palette interactions	Change	What the palette saves
Connect Azure	13	12 (mouse clicks: 4)	-1	reaches the connect dialog in 3 instead of 4 (Cloud Cost, Azure, Manage Accounts, Connect Account); the wizard's own 9 are unchanged
Connect AWS	11	10 (mouse clicks: 4)	-1	same: 3 instead of 4 to reach the dialog
Connect Google Cloud	7	7 (mouse clicks: 1)	+0	lands on the form with Display Name focused (no Manage click, no focus click)
Connect SAP BTP	9	9 (mouse clicks: 1)	+0	lands on the form with Account Name focused
Connect SAP system	7	7 (mouse clicks: 1)	+0	opens the destination dialog directly, name focused
Add budget	5	6 (mouse clicks: 1)	+1	no saving: Ctrl+K + query + Enter (3) costs more than the 2-click menu; the win is keyboard-only operation with the name field focused
Upload invoice	7 (4 to reach /invoices + 3)	5 (mouse clicks: 2)	-2	removes the hidden-page detour (the largest single saving)
Add user (default role)	7	7 (mouse clicks: 0)	+0	zero mouse clicks end to end
Upload security notes	5	5 (mouse clicks: 1)	+0	dialog opens with the textarea focused; zero mouse clicks except Upload

Reading: the palette removes navigation, not form overhead. It cuts the connect launch from 4 clicks to 3 interactions and makes every flow operable with the keyboard, but the Azure and AWS wizards still spend 4 clicks on screens with no input (Next, Validate details, Save account, Done). That is why section 8 still ranks the connect flows first.

Other improvements already visible on the live tree (measured with the same harness, section 3 has the full rows): Dashboard Quick Access 'Add SAP System' now opens System Configuration; Cost Overview's 'Connect cloud account' wizard now completes; a newly connected Azure account can now be opened.

Task	Baseline	Live tree
Add SAP system from Quick Access	8	5
Cost Overview: Connect cloud account	11 (did not complete)	12
Connect Azure, then open the new account	14 (did not complete)	14

Task	Baseline	Live tree
Ctrl+K 'connect azure' opens the dialog	2 (did not complete)	3

7. Page inventory (sections, tabs, dialogs, components)

From reading each page and its section components. 'Shared' = components from `src/shared/ui`; 'hand-written' = markup that duplicates one of them.

Page	Sections	Tabs (hash)	Dialogs, drawers	Shared used	Hand-written duplicates
Dashboard `/`	Pulse header and greeting; DashboardToolbar (search, Time range, Environment, System, Refresh); module cards; sidebar (subscription, Quick Access, support); 6 KPI glance cards; Needs attention (max 6); SAP health table; Cloud cost by provider donut.	no tabs	donut View data dialog	SearchField, ModuleCard, SapSystemsPanel, Select	KPI cards, glance strip; search and Selects do nothing
Cost Overview	Header with Cloud/Invoices toggle, Refresh, Connect cloud account; 4 KPIs; Cost at a glance (trend with period toggle, AI insights, MSP savings); spend by provider, share, movers; invoice strip; recent alerts; footer totals.	no tabs (view toggle is state)	ConnectCloudAccount Wizard; View data dialogs	SegmentedToggle, EmptyState, InsightList, PageHeader	6 `<a href>` links, KPI shells
Cost Analytics	Header (Save view, Export report); scope filter bar (Providers, Range, Group by, saved views); KPIs; monthly line chart; breakdown table + bar list + selection card.	none	none	PageHeader, Select, BarList	filter bar, saved-view chips
Savings & Optimization	Provider Select; TabBar Savings overview (#overview) / Opportunities (#opportunities): KPIs, trend + category donut, top recommendations, MSP table; filters, table, saving-by-type.	#overview, #opportunities	opportunity drawer (Owner, dismiss reason, Start review); category drawer	TabBar, DetailDrawer, SegmentedToggle, Select, EmptyState, BarList	filter bar
Reserved Instances	Provider Select; KPIs; reservations table; utilisation chart; commitment recommendations.	none	none	PageHeader, Select	table, chart without data table
Resource Inventory	Export CSV; KPIs; FilterBar (search + 5 Selects + Clear); table.	none	resource drawer	FilterBar, SearchField, DetailDrawer, CopyButton, Select	none
Budgets & Alerts	TabBar Budgets (#budgets: create form + list) / Alert integrations (#integrations: channel list).	#budgets, #integrations	AddChannelDialog; confirms for delete budget and remove channel	TabBar, ConfirmProvider, useModalOverlay, Select	`div.mv-empty`
Smart Alerts	TabBar Alerts / Recipients / Settings; Alerts: summary line, Mark All Read, Refresh, KPIs, filters (status, type, provider), table with Pagination.	#alerts, #recipients, #settings	alert detail drawer	TabBar, DetailDrawer, Pagination, SegmentedToggle, Select	filter container
Alerts `/alerts`	Header and one static table of 3 rows.	none	none	PageHeader	table
Invoices	Header (Refresh, Upload Invoice); 6 KPIs; split panel: provider toggle with counts + invoice cards + Pagination; analysis panel.	none	UploadModal; delete confirm	SegmentedToggle, Select, Pagination, ConfirmProvider	modal, cards, KPIs, empty markup

Page	Sections	Tabs (hash)	Dialogs, drawers	Shared used	Hand-written duplicates
Billing	Split panel: your invoices; invoice detail with Pay Now.	none	none	PageHeader	list, empty states
AI Assistant page	Header; conversation with suggestion chips; composer.	none	none	PageHeader	conversation duplicates the widget
Azure overview / manager	Overview: heading, Refresh Data, Manage Accounts, KPIs, Environments table, trend, top services. Manager: account list (rename, details, delete), limit badge, Connect Account.	local view switch (dashboard/ accounts)	connect wizard (4 steps); details dialog; delete confirm	ClickableRow, RowOpenButton, ConfirmProvider, SpendTrendChart	wizard, dialogs, Sparkline
Azure / AWS / GCP / BTP account	Header (Back, switch account, month, AI Insights, Sync); cost-view nav (Executive, Operations, Technical; BTP adds Hierarchy); CloudSpendSummary; per-view panels.	view switcher (state, not hash)	View data dialogs; BTP hierarchy drill-down	CloudSpendSummary, ServiceBreakdownList, ServiceCategorySummary, DailyCostHistory, CostByRegion, SegmentedToggle, DetailGroup	cost tabs, dropdowns, tables (x4)
AWS / GCP / BTP overview and managers	AWS mirrors Azure (overview, manager, wizard). GCP: overview, manager with inline form. BTP: overview, manager with inline 5-field form, service-instances table.	none	AWS wizard; GCP and BTP none	ClickableRow (AWS), ConfirmProvider	rows, forms
SAP Overview	Destination switcher; 4 KPIs; load chart + condition donut; Needs attention + unread by severity; systems table.	none	View data dialogs	PageHeader, BarSeriesChart, DonutChart, InsightList, BarList	condition pills, table
HANA/DB Monitoring	Host panel (Refresh, SyncStatus); 4 KPIs; TabBar Vitals / Services and disk / Backups.	#vitals, #storage, #backups	View data dialog (3 copies)	TabBar, useHashTab, InsightList	SegmentedSwitch, dialogs
Performance	Status panel; 8 KPIs; TabBar System health / Jobs and processes / Analytics.	#health, #jobs, #analytics	View data dialog	TabBar, useHashTab	SegmentedSwitch, dialog
Jobs & Dumps	TabBar Background jobs / Short dumps; FilterBar with SearchField and FilterChips; table with Show more.	#jobs, #dumps	job drawer; dump drawer	TabBar, FilterBar, SearchField, FilterChips, DetailDrawer, EmptyState, CopyButton	inline empty block
Alerts & Escalations (SAP)	TabBar Alerts / Settings / Recipients; source toggle, KPIs, filter bar, table with Pagination; settings forms; recipient form.	#alerts, #settings, #recipients	none	TabBar, SegmentedToggle, Select, Pagination	filter bar
Transport Management	KPIs; import route form; filter bar (status toggle, search, Import Selected, Export); table; activity log; import queue.	none	ActivityLogDrawer	FilterBar, SearchField, SegmentedToggle, Pagination, DetailDrawer	none
Security Notes	Header (History, Check Status, Upload); 6 KPI cards; FilterBar with 3 Selects; table with Pagination.	none	UploadNotesDialog; NotesHistoryDrawer; delete confirm	SearchField, FilterBar, Select, Pagination, DetailDrawer, ConfirmProvider	upload modal shell

Page	Sections	Tabs (hash)	Dialogs, drawers	Shared used	Hand-written duplicates
BTP Security	Header Refresh (dead); tab strip Users / Role Collections / Roles; search and status filter; empty tables.	state only	none	SearchField, Select	tab strip, filter bar
System Configuration	Add Destination; status line + search; destination cards (Connect/Disconnect, edit, remove).	none	DestinationDialog (add and edit); remove confirm	SearchField, Select, SegmentedToggle, PasswordField, ConfirmProvider	cards, empty block
User Management	Header (search, Add user); seat summary; TabBar Members / Roles / Permissions; AddUserDialog; RoleDialog; RoleDetailsDrawer; PermissionExplorer with unsaved-changes bar.	#members, #roles, #permissions	AddUserDialog; RoleDialog; role drawer; 3 confirms	SearchField, TabBar, EmptyState, Pagination, Select, DetailDrawer, ConfirmProvider	Toggle, StatusBadge, rows
Audit Trail	Header (Export CSV, Manual re-sync); FilterBar (search, actor radiogroup, module Select, date range); day-grouped timeline; Pagination.	none	AuditEventDrawer	FilterBar, SearchField, Select, Pagination, DetailDrawer	actor radiogroup
Sync Logs	Header Refresh; Status and Provider toolbar; table; Pagination.	none	log drawer	Pagination, DetailDrawer, EmptyState	toolbar, no search
Subscription	TabBar Your plan / Available plans.	#plan, #plans	PlanRequestDialog	TabBar, PageHeader	rows
Account	TabBar Profile / Security / Appearance / Help; change-password form.	#profile, #security, #appearance, #help	none	TabBar, PasswordField	theme and density radiogroups
Super admin: dashboard, organizations, editor, billing, generator, customers	Overview with two ModuleCards and two tables; organizations table and form with sticky action bar; billing split panel with Invoices / Payment Terms; invoice generator with live preview; customers form and list.	Invoices/Payment Terms (state)	send-invoice and sent modals; discard confirm	ModuleCard, PageHeader, Select, ConfirmProvider	tables, search, modals, window.prompt

Shared component adoption

Shared component	Used by	Hand-written copies instead
SearchField	Dashboard, Resources, Jobs, Transport, Security Notes, Configuration, Users, Audit, BTP Security	Admin Billing own search input; Sync Logs, Smart Alerts, Alerts, Cost Analytics, Optimization, providers have none
FilterBar	Resources, Jobs, Transport, Security Notes, Audit	`mv-filter-bar` divs in Cost Analytics, Optimization, SAP Alerts, BTP Security; `mv-cost-alert-filters` in Smart Alerts; Sync Logs toolbar
FilterChips	Jobs only	Smart Alerts status/type (SegmentedToggle without counts), SAP Alerts (Selects), Transport status, Audit actor radiogroup, Sync Logs buttons, saved views in Cost Analytics, SAP condition pills

Shared component	Used by	Hand-written copies instead
TabBar + useHashTab	Budgets, Smart Alerts, Optimization, HANA, Performance, Jobs, SAP Alerts, Users, Subscription, Account	4 provider cost-view navs, BTP Security strip, Admin Billing pair, notification drawer All/Unread, Cost Overview Cloud/Invoices (SegmentedToggle)
ClickableRow / RowOpenButton	Azure and AWS overview tables	Every other row-opening table (Smart Alerts, Audit, Sync Logs, roles, invoices, systems, jobs, resources); Admin Billing rows are `tr onClick` only
EmptyState	Cost Overview, Jobs dumps, Users, Audit, Sync Logs, Invoices summary	`mv-empty`, `mv-empty-inline`, plain `<p>` on about 20 pages (see section 5)
useModalOverlay (Escape, trap, focus return)	Add user, role, plan request, destination, upload notes, add channel, search, bell, DetailDrawer	ConfirmProvider (every confirm), wizard, invoice UploadModal, SendInvoice and SentEmail modals, 3 chart View data dialogs, super-admin nav drawer
SegmentedToggle	Cost Overview, Daily cost history, Invoices, SAP alerts source, Transport, Smart Alerts	SAP SegmentedSwitch, Account Theme/Density radios, Audit actor group
ChartDataDialog (View data)	Daily cost history, Cost by region, BarSeriesChart, BarLineChart	DonutChart, SpendTrendChart, CumulativeSpendChart, SAP ChartDataDialog, HanaVitalsChart, Azure ReservationUtilizationChart

8. Ranked friction list (top 25)

Ranked by measured interactions weighted by how often the task is done and by the owner's priority (new organisations connecting accounts first). 'Before' is the measured baseline from the Dashboard; 'After' is the target for the change named; files are the ones to touch. Palette numbers are from section 6.

#	Task	Before (measured)	Change that reduces it	After target	Files
1	Connect Azure	13 (8 clicks + 5 fields; palette 12)	One shared ConnectAccountDialog: opened from the Dashboard, empty states and Ctrl+K; 'Use sample credentials' fills valid values; fold Validate into Connect and close on success (no Done); open the new account	3 with sample values (launch, Use sample, Connect); 7 with real values (launch + 5 fields + Connect)	features/cloud/components/ConnectCloudAccountWizard.tsx; features/azure/pages/AzureAccountManager.tsx, AzureOverallDashboard.tsx, AzureRoot.tsx; features/dashboard/pages/HomePage.tsx; app/layouts/pageSearch.ts
2	Connect AWS	11 (8 clicks + 3 fields; palette 10)	Same dialog; ask Region as a default Select; sample = AKIAIOSFODNN7EXAMPLE pair	3 with sample; 5 with real values	features/cloud/components/ConnectCloudAccountWizard.tsx; features/aws/pages/AwsAccountManager.tsx, model/AwsAccountContext.tsx
3	Connect SAP BTP	9 (4 clicks + 5 fields; palette 9)	Same dialog; auto-derive the Reporting URL from the Token URL; success toast; fix the saved region (`reporting` instead of `eu10`)	3 with sample; 6 with real values	features/btp/pages/BtpAccountManager.tsx, model/BtpAccountContext.tsx
4	Connect Google Cloud	7 (4 clicks + 3 fields; palette 7)	Same dialog; paste the service-account JSON and derive Display Name and Project ID from it (1 field instead of 3)	3 with sample; 3 with a pasted key	features/gcp/pages/GcpAccountManager.tsx, model/GcpAccountContext.tsx
5	Add SAP system / destination	7 via the menu; Quick Access 8 at baseline (5 live); palette 7	Dialog with a 'Use sample destination' button and a real test-connection state; SID defaults from the URL host; make the destination switcher read the destinations list	3 with sample	features/sap/components/DestinationDialog.tsx, pages/SystemConfigurationPage.tsx, components/DestinationSwitcher.tsx; features/dashboard/components/DashboardSidebar.tsx
6	Upload an invoice (reach + upload)	7 (reach 4 + upload 3); palette 5	Put Invoices in the Cloud Cost menu for every org that has the module; palette 'Upload invoice' opens the file chooser; accept a dropped file anywhere on the page; detect the provider from the file name	3 (menu 2, file chooser, auto-submit)	app/router/navConfig.ts; features/invoices/pages/InvoicesPage.tsx; app/layouts/pageSearch.ts
7	Delete an invoice (reach + delete)	7 (reach 4 + delete 3)	Menu entry (above) and a row trash action with an Undo toast in place of the confirm dialog	3 (menu 2 + trash, undo toast)	features/invoices/pages/InvoicesPage.tsx; shared/ui/ConfirmProvider.tsx
8	Edit permissions of an existing custom role	7 (7 clicks)	Role row and drawer 'Edit permissions' open the matrix with that role selected and its first area open; remember the last tab; keep the draft when switching tabs	4 (menu 2, row action, switch + Save)	features/users/pages/UserManagementPage.tsx; features/access/components (PermissionExplorer)
9	Add a user with a chosen role	8 (6 clicks + 2 fields); default role 7; palette 7	Role chips in the dialog (remember the last role), pre-tick provider access from the role, add-another	5 with a chosen role; 4 default	features/users/components (AddUserDialog)

#	Task	Before (measured)	Change that reduces it	After target	Files
10	Security notes: upload then resolve status	6 (5 clicks + 1 field); upload only 5; palette upload 5	Run Check Status automatically after Upload (checkbox, on by default); bulk delete	3 on the page (Upload, numbers, Upload; Check Status runs by itself)	features/sap/pages/SecurityNotesPage.tsx, components (UploadNotesDialog), model
11	Add a custom role and give it permissions	8 (6 clicks + 1 field + 1 key)	'Start from' defaults to a copy of the Analyst role; the role dialog saves and opens the matrix on the right module	5	features/users/pages/UserManagementPage.tsx; features/users/components (RoleDialog)
12	Pay an invoice	4 (4 clicks)	Pre-select the first unpaid invoice; show Pay Now on the list row	3 (menu 2 + Pay Now)	features/billing/pages/BillingPage.tsx, components
13	Add a notification channel	6 (5 clicks + 1 field)	Default recipient = signed-in user's e-mail; Add channel in the page header on both tabs	4	features/budgets/pages/BudgetsPage.tsx, components/AddChannelDialog.tsx
14	Create a budget	5 (3 clicks + 2 fields); palette 6	Suggest a name and a limit from the spend trend; submit with Enter; one entry point (Cost Overview 'create one' link is a hard reload)	4 (menu 2, amount, Enter)	features/budgets/pages/BudgetsPage.tsx; features/cloud/pages/CostOverviewPage.tsx
15	Open a provider account detail	3	Single connected account opens directly from the menu item; 'last viewed account' tile on the Dashboard	2 (1 from the Dashboard tile)	features/azure/pages/AzureRoot.tsx (and AWS, GCP, BTP roots); features/dashboard/pages/HomePage.tsx
16	Switch to Operations or Technical view	4	Shared CostViewTabs on TabBar + useHashTab (+ remembered per account) replaces 4 copied navs	3 the first time, 0 extra afterwards	features/azure/pages/AzureCostPage.tsx, aws/pages/AwsCostPage.tsx, gcp/pages/GcpCostPage.tsx, btp/pages/BtpCostPage.tsx; shared/ui/TabBar.tsx
17	Start a guided tour	first time 2; returning 3	Visible 'Tour' button in every page header and a palette 'Take the tour' action	1	features/assistant/components/GuidedTour.tsx, AssistantWidget.tsx; app/layouts/pageSearch.ts
18	Resolve a cost alert from the Dashboard	2 (lands on the list, not the alert)	Resolve and Ignore inline on 'Needs attention' rows; unify the five unconnected alert sources	1	features/dashboard/pages/HomePage.tsx, model/useDashboardData.ts; data-access/repositories/smartAlertsRepository.ts
19	Change the spend period on Cost Overview	3	Remember the period (useRememberedState) and put it in the URL	3 the first time, 2 afterwards (menu only)	features/cloud/pages/CostOverviewPage.tsx
20	Reach /alerts, /invoices, /ai-assistant	/alerts 1 (Dashboard link only), /invoices 4, /ai-assistant 4	Add the three to their natural menus (Alerts under Cloud Cost > Governance, Invoices, AI Assistant) and to the palette results	2 each	app/router/navConfig.ts, route-manifest.ts; app/layouts/pageSearch.ts

#	Task	Before (measured)	Change that reduces it	After target	Files
21	Open the critical system from SAP Overview	not possible (dead end)	Make system rows ClickableRow: open the HANA page for that system or a detail drawer; pills become FilterChips	3 (menu 2 + row)	features/sap/pages/SapOverviewPage.tsx, components/SapSystemsPanel.tsx
22	Open HANA backup details	4	Remember the HANA tab; make backup markers keyboard-focusable buttons that open the detail	3 (menu 2 + marker) when #backups is remembered	features/sap/pages/HanaMonitoringPage.tsx, components (HanaBackupTimeline)
23	Delete a security note	4 (4 clicks)	Row trash with an Undo toast instead of a confirm; bulk select + delete	3	features/sap/pages/SecurityNotesPage.tsx; shared/ui/SelectionBar.tsx
24	Request a plan	5	Show plans on the same tab as the current plan; Request plan from the profile menu	4	features/subscription/pages/SubscriptionPage.tsx
25	Forgot password and other public-page reach	/forgot-password 3; /register has no inbound link	Link /register from /company-signup and /login help text (sign-in flow itself is out of scope, so only add the register link on /company-signup)	2	features/auth/pages/CompanySignupPage.tsx, RegisterPage.tsx

Cross-cutting fixes that remove friction on many pages at once (not counted as separate rows above):

- Make ConfirmProvider and every hand-written modal use `useModalOverlay`: Escape, focus trap and focus return are missing on all confirms, the invoice upload modal, the connect wizard, the invoice-send modals and 4 chart dialogs (`shared/ui/ConfirmProvider.tsx`, `ConnectCloudAccountWizard.tsx`, `InvoicesPage.tsx`, `BarSeriesChart`, `DonutChart`, `SpendTrendChart`).
- Replace the 6 plain `<a href>` links on Cost Overview with `Link`: today each one reloads the app and wipes the in-memory demo state (`features/cloud/pages/CostOverviewPage.tsx`).
- Move the floating assistant launcher out of the way of drawer footers: it covers the primary button of the Optimization drawer (and any right-aligned drawer action); measured, the click had to be placed on the button's left edge.
- Inner scrollbars: 3 tables scroll vertically inside a wrapper at 1440 (Jobs, Cost Analytics, Resource Inventory) and 12 scroll horizontally at 390 (SAP Overview, Performance, Jobs, BTP Security, Azure, AWS, BTP, Optimization, Billing, Users, Sync Logs, Alerts); the owner rule is no inner scrollbars (a stacked or card layout below the breakpoint, no max-height on the wrapper).
- Remove or wire the controls that look interactive but do nothing: SAP BTP Security (dead Refresh, empty tables), BTP service-instances Refresh, the Dashboard search box and its 3 Selects, `/alerts` (features/sap/pages/BtpSecurityPage.tsx, features/btp/pages/BtpServiceInstancesPage.tsx, features/dashboard/components/DashboardToolbar.tsx).
- Persist state in the URL: add the filter/period/view keys to `useHashTab`-style hooks so a link restores them (Cost Overview, Smart Alerts, SAP Alerts, Jobs chips, Audit, provider views).

9. Not automated, caveats, and how to rerun

- **Reach counts include no scrolling or waiting.** Menus open on click (hover never opens them), so 2 is the true minimum.
- **Picking a file in the OS dialog is not counted** (Playwright sets the file); the click that opens the picker is. A real user adds a few seconds, not a click.
- **Demo data limits.** Zero-account states cannot be reached with the six demo accounts (all have connected accounts), so empty-state verdicts are from code. The new empty-organisation account in the plan does not exist yet.
- **Estimated, not measured**: register steps, company-signup full flow, Reset password, super-admin organisation settings save, customer add, theme change on the page, composer typing on `/ai-assistant`, the Performance chip link. Each is labelled 'code count' in section 5.
- **Defects taken from code reading** and not re-verified in the browser: BTP region saved as `reporting`, AWS region hard-coded, GCP flow-guide targets that do not exist, the demo store lost when leaving `/aws`, `/gcp`, `/btp`. The Azure 'cannot open the new account' defect was measured (task `connect.azure-open`).
- **The Cost Overview wizard** did not finish saving at baseline (still 'Saving...' after 2.5 s: it posts to a real API); on the live tree it completes.
- **Moving target.** The baseline is commit `9a087dc`, served from a `git archive` copy so it is reproducible. The live tree differed from it by 114 modified files while this audit ran (command palette actions, provider account stores, assistant widget), so a handful of live numbers changed between my own two live runs; the table in section 3 shows the baseline and, after `/ now`, the live value from the last run.
- **Sign-in is untouched.** The harness uses the Demo mode card button for every session (1 click, not counted).

Rerun: from `D:\mavera` with the dev server on :5173, `python tools/visual/click_audit.py --json audit.json --md audit.md` (about 25 minutes for everything). Useful flags: `--only <id or group>` (for example `--only connect`, `--only palette`), `--list`, `--skip-layout`, `--layout-only`, `--keyboard-only`, `--extra-layout`, `--shots <dir>` (screenshot of each failed task), `--headed`. `BASE_URL` points it at another origin (for example a `git archive` copy of a commit served on another port, which is how the baseline here was pinned).